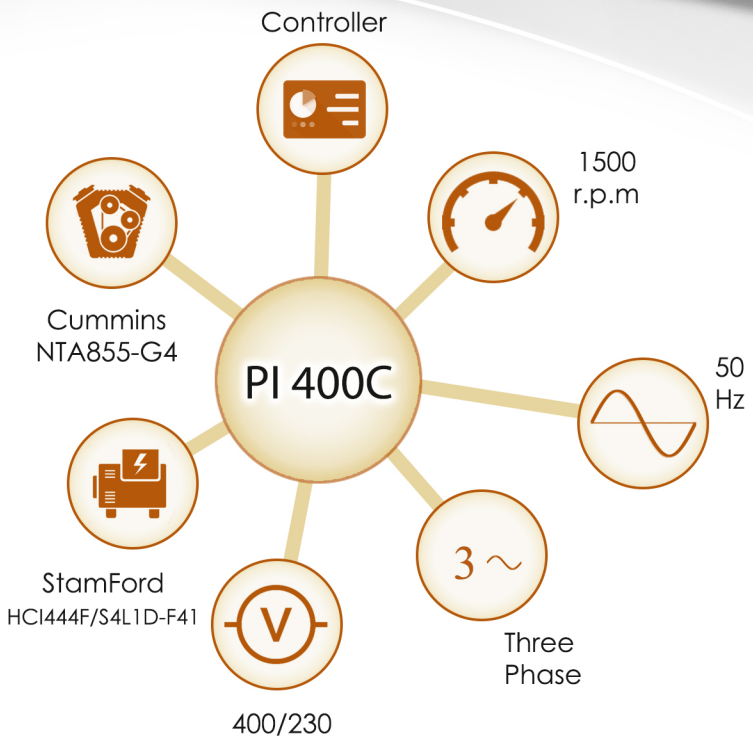

TUV NORD

OHSAS 18001 — ISO 14001 — ISO 9001

UAE
صنع في الإمارات
MADE IN UAE

Mechancial Power driven by Cummins

- Manufactured in facilities certified with ISO 9001:2015, ISO 14001:2015 & OHSAS 18001:2007.
- Manufactured in accordance to 8528-1 to 12.
- Engine performance according to ISO 3046, BS 5514, DIN 6271.
- Alternator performance according to NEMA-MG1, BS 5000, DIN EN, relevant ISO, IEC60034.
- Breaker complise with IEC – 60947-2.

الصناعات الدقيقة
PRECISION INDUSTRIES

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info@pi-dubai.com | www.pi-dubai.com

الصناعات الدقيقة
PRECISION INDUSTRIES

PI 400C

Industrial Generating Set

POWERED BY Cummins

MODEL	rpm / Hz	VOLTAGE	PRIME (1)	STANDBY (2)
PI 400C	1500 / 50	400 / 230	300.0 kVA / 240.0 kWe	330.0 kVA / 264.0 kWe

ENGINE SPECIFICATIONS	
Rated Output (PRP) (1)	317 kW_m
Rated Output (ESP) (2)	351 kW_m
Engine Make & Model	Cummins NTA855-G4
No. of Cylinders	6 Vertical In-line
Cycle	4 Strokes
Aspiration	Turbocharged and Aftercooled
Cooling Method	Water
Governing Type	Electronic
Governing Class	G2 - ISO 8528 Part 1
Compression Ratio	14.0 : 1.0
Displacement	14.0 L / 855 in^3
Bore/Stroke (mm / in)	(140/152) / (5.5/6.0)
Battery and Charger Alternator	24 VDC, 35 Amp

AIR SYSTEM	
Air Filter Type	Dry Element
Combustion Air Flow (PRP)	24.48 m^3/min
Combustion Air Flow (ESP)	26.04 m^3/min
Radiator Air Flow	456 m^3/min

COOLING SYSTEM	
Total Coolant Capacity	20.8 L / 5.5 US gal
Water Pump Type	Centrifugal Eng-Driven
Radiator Fan Load	11.6 kW
Heat Radiation to Room (PRP)	46 Kw
Heat Radiation to Room (ESP)	51 kW

LUBRICATION SYSTEM	
Oil Filter Type	Spin on full flow filter
Total Oil Capacity	38.6 L / 10.2 US gal.
Oil Pan	36.0 L / 9.5 US gal.
Oil Type	API CH4/CI4; SAE 15W-40

FUEL SYSTEM	
Fuel Filter: Spin on full flow filter with water separator	
Recommended Fuel	Class A2 Diesel
Fuel Consumption Standby	84.0 L/hr / 22.3 US gal/hr
Fuel Consumption 100% PRP	76.0 L/hr / 20 US gal/hr
Fuel Consumption 75% PRP	57.0 L/hr / 15 US gal/hr
Fuel Consumption 50% PRP	39.0 L/hr / 10.2 US gal/hr

EXHAUST SYSTEM	
Muffler Type	Industrial Grade
Max. Back Pressure	10.13 kPa
Exhaust Gas Flow (PRP/ESP)	67.68 / 73.50 m^3/min
Exhaust Gas Temperature (PRP/ESP)	524 / 541 ˚C

ALTERNATOR SPECIFICATIONS	
Rated Output (Prime) (1)	400 kVA
Rated Output (Stand by) (2)	450 kVA
Alternator Make & Model	Stamford HCI 444F/ S4L1D-F41
Number of Poles	4
Number of Winding Leads	12
Type of Bearing	Single
Insulation Class / Temp Rise	H / H
Efficiency	92.8%
Ingress Protection Rating	IP 23
Excitation System	Self Excited
AVR Model	Stamford - AS440

ALTERNATOR OPERATING DATA	
Overspeed	2250 r.p.m
Voltage Regulation	± 1.0 %
Wafeform distortion	No load <1.5% Linear load <5%
Radio Interface	Standard EN61000-6-2:2001
Cooling Air Flow	0.8 m^3/sec

(1) **PRIME POWER RATING (PRP):** PRP is defined as the maximum power which a Generating set is capable of delivering continuously whilst supplying a variable electrical load when operated for an unlimited number of hours per year. The permissible average power output over 24 hours shall not exceed 70% of PRP unless otherwise agreed by RIC engine manufacturer. An overload capability of 10% of 100% of the prime rated electrical power is permitted for emergency use for a period of 1 hour within 12 hours of operation

(2) **EMERGENCY STANDBY POWER RATING (ESP):** ESP is defined as the maximum power available during a variable electrical power sequence, under the stated operation condition, for which a generating set is capable of delivering power in the event of a utility power outage or under test condition for up to 200 Hours of operation per year. The permissible average output over 24 hour of operation shall not exceed 70 % of the ESP power rating noting that no over load is permitted.

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Data tolerance limit +/-5%

ISO 9001

PI 400C

Industrial Generating Set

POWERED BY Cummins

CONTROLLER SPECIFICATIONS

Controller Make & Model	DeepSea 6120
Operation Mode	MRS / AMF (optional)
Display	Graphic Back-lit LCD (128x64) pixles
Ingress Protection Rating	IP65
Binary Inputs/Outputs	6 / 4
Analog Inputs	4
Measurement	Vac, A, Hz, kVA, kW, Vdc
Event Log	Alarms log, Hrs log
Communication	USB

ENCLOSURE SPECIFICATIONS

Enclosure Type	Acoustic & Weather Proof
Anticorrosive Protection	Polyester Powder Coated Galvanized Sheet
Ingress Protection Rating	IP23
Lifting	ISO Standard Lifting
Emergency	External Emergency Push Button
Canopy RAL Color	RAL 2000
Baseframe RAL Color	RAL 9011
Noise Pressure level @ 7m	82.2 dB(A)

GENSET DIMENSIONS & WEIGHT

GENSET TYPE	Length (mm)	Width (mm)	Height (mm)	Fuel Tank Capacity (L)	Dry Weight (kg)	Wet Weight (kg)
OPEN	3127	1145	2182	695	3500	3550
CLOSE	5205	1624	2605	1030	4140	4190

STANDARD MECHANICAL FEATURES

- Genset design provides a low noise level with an optimized performance of the ventilation and exhaust systems at 50 ºC ambient temperature.
- Robust structure design of Enclosure and Baseframe.
- Hevy duty lifting lugs.
- Multi doors for easy access & maintenance.
- Ingress Protection Rating according to BS EN 60529.
- Heavy Duty Baseframe with built-in tank & forklift pockets.
- Industrial Grade Muffler with rain cap.

STANDARD ELECTRICAL FEATURES

- An advance Control system is designed to provide a comperhensive protection and to monitor the parameters of generating set.
- MCCB power circuit breaker.
- Battery with charging alternator, cables, and tray.
- Sealed harness & high resistant electrical connections.
- Fast and accurate protection response.
- Generating Set remote start function.
- Numeric display with LED. Various languages capable.

OPTIONAL FEATURES

- Advanced Controllers are available on request.
- 4 poles manual / Motorized Circuit breaker
- Jacket water pre-heater
- Static Battery Charger
- Residential / Critical grade muffler
- Fuel Filter / Water seperator Fuel Filter
- Remote Annunciator

Application

Infrastructure, Industrial, Residential, Telecom, Defence, Mining, Agriculture,

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